

Harshil Makwana

Generative AI Engineer | AI Automation & Agentic Systems

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SUMMARY

Computer Engineering graduate (CGPA 8.39) with hands-on experience building LLM-powered autonomous agents, RAG pipelines, and generative AI systems across two internships — shipping 10+ production AI workflows and 5+ live public projects, including dual-platform publishing systems and REST APIs. Skilled in integrating LLMs with vector databases, async task pipelines, and multi-platform APIs to deliver reliable, production-grade automation. SAP Code Unnati certified (AI, Machine Learning, Deep Learning, IoT, SAP BTP) via Edunet Foundation. Focused on retrieval-augmented generation, multi-step agent orchestration, and tool-integrated LLM pipelines from prototype to cloud deployment.

TECHNICAL SKILLS

Generative AI & LLMs: LLMs, Prompt Engineering, OpenAI API (GPT-4o, GPT-image), RAG Pipelines, Embeddings, Semantic Search, NLP, Hugging Face Transformers

Agentic Systems: Autonomous Task Agents, Multi-step Tool Orchestration, Self-Evaluation/Scoring Loops, n8n Agentic Workflows

RAG & Retrieval: Context-Injected Prompting, Chroma DB, FAISS, Semantic Search, Embedding Based Retrieval

Backend & Deployment: Python, FastAPI, REST APIs, Celery (Async Task Queues), Docker, Render, Railway, Netlify, Git, Linux

ML & Data: PyTorch, TensorFlow, Scikit-learn, Bidirectional LSTM, Neural Networks

Integrations: Twilio WhatsApp, Shopify GraphQL APIs Instagram Graph API, WooCommerce API, Cloudflare, Discord API, Trello API

PROFESSIONAL EXPERIENCE

AI Automation Engineer (Intern)

May 2026 – Present

Rohitre Media — Remote

- Built a Telegram-driven publishing agent that converts a headline and source image into a branded graphic, drafts and reviews the caption and website summary, and publishes to the company website and Instagram with independent per-platform state tracking, so a partial failure retries only the failed side.
- Extended a content-generation agent to auto-generate and host thumbnails and publish end-to-end with zero manual writing, running 24/7 with Telegram alerts; improved publishing efficiency by ~95%, cut writer time 5x, and maintained ~90% accuracy on spot-checked output.
- Designed an autonomous content agent that scans 20+ YouTube channels, filters relevant sports content, and drafts articles using a self-evaluation loop that scores each draft against an SEO rubric and regenerates up to 2 revisions before selecting the best version.
- Built a Python-based Telegram agent for a Shopify store that generates AI model photos, drafts titles, descriptions, pricing and captions, and publishes after owner approval, with independent Shopify and Instagram/Facebook publish tracking and chat-based product management, cutting manual upload effort ~85% and publishing time ~90%.

Python AI Engineer (Intern)

Nov 2025 – May 2026

Toshal Infotech Pvt. Ltd. — Surat, India

- Engineered AI-powered FastAPI microservices with Celery async task pipelines, cutting data-retrieval latency; implemented a PGVector-based RAG pipeline enabling sub-second semantic search over large document corpora.
- Integrated OpenAI LLMs for automated insight generation (summaries, keyword extraction, sentiment, topic detection) and channel tracking alerts, reducing manual analysis time significantly; deployed and maintained services on Render.

KEY PROJECTS

Full Agentic Controlled Shopify Store

Python · Telegram Bot API · Shopify GraphQL API · OpenAI API · Instagram & Facebook Graph API

- Built a Telegram agent that generates AI model photos for Shopify products, drafts titles, descriptions, pricing and captions, and publishes only after owner approval, using a stateful category-based intake flow with a 24-hour approval gate.
- Added independent Shopify and Instagram/Facebook publish tracking so a failure retries only the failed side, plus chat-based product management (lookup, out-of-stock sizes, delete) via Shopify APIs

AI-Powered Video Intelligence Platform

FastAPI · PostgreSQL · PGVector · Celery · OpenAI API · Render

- Architected a full-stack LLM + RAG platform over video transcripts with ~90% retrieval accuracy via semantic search, plus automated alerts and new-upload channel monitoring, deployed on Render as a scalable, production-ready service.

EDUCATION

Vidhyadeep Institute of Engineering and Technology, GTU, Gujarat, India

B.E., Computer Engineering — CGPA : 8.39

Graduated May 2026

Bhagwan Mahavir University

Diploma, Computer Engineering — CGPA : 8.21

2020 – 2023

CERTIFICATIONS & ACHIEVEMENTS

- SAP Code Unnati Certification – Edunet Foundation (AI, Machine Learning, Deep Learning, IoT, SAP BTP)
- 10+ production AI workflows and 5 live public projects deployed across two internships, including REST APIs, autonomous agents, and dual-platform publishing systems.